

Non-Random Missingness in Anthropometric Measurements: A Multi-Country Analysis of South Asian DHS Data (2011–2022)

Tahsina Akter
SK-182-033

Institute of Statistical Research and Training
University of Dhaka

March 09, 2026

- Introduction
- Literature Gap and Study Rationale
- Research Question and Objective
- Data and Variables
- Analytical Framework and Key Findings
- Conclusions
- Limitations and Future Directions

- Anthropometric measurements are standardized body measurements used to assess child growth and nutritional status.
- Height-for-age Z-score (HAZ) is a key indicator of chronic undernutrition and stunting among children under five.
- In large-scale surveys, some child height observations are missing or biologically implausible, reducing data usability.
- If unusable observations are systematically patterned, complete-case analysis may bias inference.

Literature Gap and Study Rationale

- HAZ is widely used in research on child growth, undernutrition, and inequality.
- Missing or implausible anthropometric values are often excluded in empirical studies.
- This practice may assume that unusable observations are not systematically patterned.
- In South Asian DHS data, evidence remains limited on whether anthropometric usability varies across social groups and survey rounds.
- This study examines whether unusable HAZ observations are systematically associated with observable characteristics and whether this affects substantive inference.

Main question:

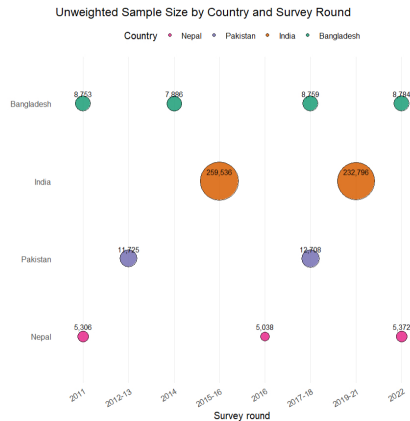
Are missing or biologically implausible child height observations in pooled South Asian DHS data random, or are they systematically associated with observed demographic and socioeconomic characteristics?

Study objectives:

- Estimate the prevalence of unusable height observations in pooled DHS surveys.
- Identify predictors of height usability using survey-weighted logistic regression.
- Assess whether usability patterns changed over time.
- Compare complete-case and multiple imputation estimates for HAZ determinants.

Data Source and Study Sample

- 11 DHS survey rounds conducted between 2011 and 2022.
- Countries included: Bangladesh, India, Pakistan, and Nepal.
- Data source: DHS Children's Recode (KR) files for children aged 0–59 months.
- Pooled unweighted sample: 566,663 children.



Unweighted sample size by country and survey round

Outcome and Key Explanatory Variables

Outcome measures:

- **Height usability** (binary): 1 = observed and plausible HAZ; 0 = missing or implausible HAZ.
- **HAZ** (continuous): used in substantive models of child nutritional status.

Key explanatory variables:

Child-level

- Age (months)
- Female child
- Illness in past 2 weeks

Maternal

- Mother's age
- Mother's education
- Currently pregnant

Household-level

- Wealth quintile
- Rural residence
- Female household head
- Under-5 children in household

Analytical Framework

- Survey-weighted descriptive and bivariate analysis of height usability
- Survey-weighted logistic regression for predictors of usable height data
- Interaction analysis by survey period and key socioeconomic factors
- Complete-case and multiple imputation comparison for HAZ models

Why this sequence?

It first establishes whether unusable observations are patterned, then evaluates whether that pattern materially affects substantive HAZ inference.

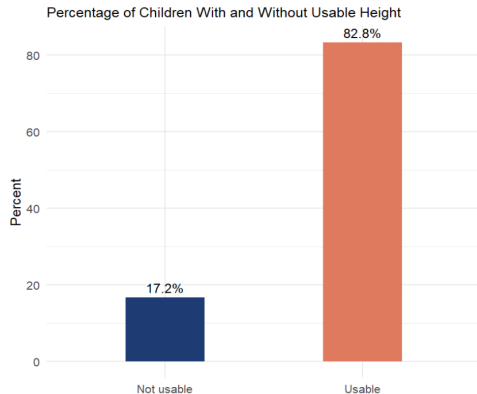
Note: Analytic sample sizes vary by model because of survey declaration, covariate completeness, and imputation.

Result 1: Overall prevalence of height usability

- Usable observations: **82.8%**
- Unusable observations: **17.2%**
- The scale of data loss is analytically non-trivial

Why it matters

This result motivates the rest of the analysis: a sizable share of children is excluded unless usability is examined carefully.



Result 2: Age is the strongest determinant of usability

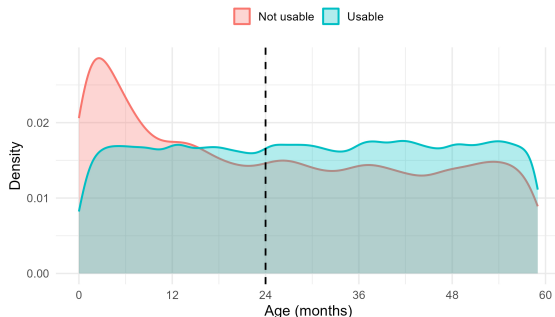
- Unusable measurements are disproportionately concentrated among younger children
- The sharpest contrast appears below age 24 months
- This pattern is consistent with the greater difficulty of measuring infants and very young children

Interpretation

The age gradient is the most robust and persistent result in the study.

Child Age Distribution by Height Usability

Dashed line indicates 24 months

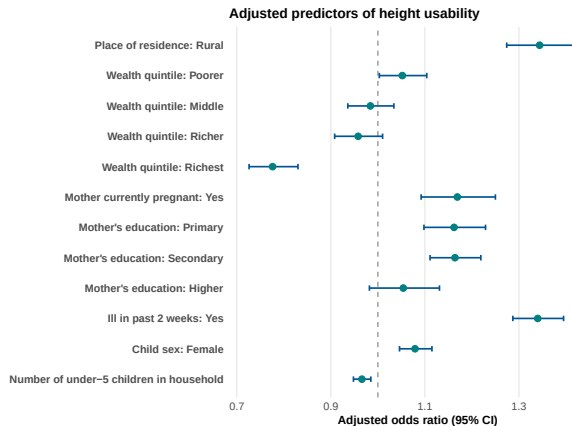


Result 3: Adjusted determinants of height usability

- Maternal primary and secondary education were associated with higher odds of usable measurement
- Rural residence showed higher odds of usable height than urban residence
- The richest households had **lower** odds of usable height than the poorest

Interpretation

Usability is not only age-related; it also follows observable socioeconomic patterns.

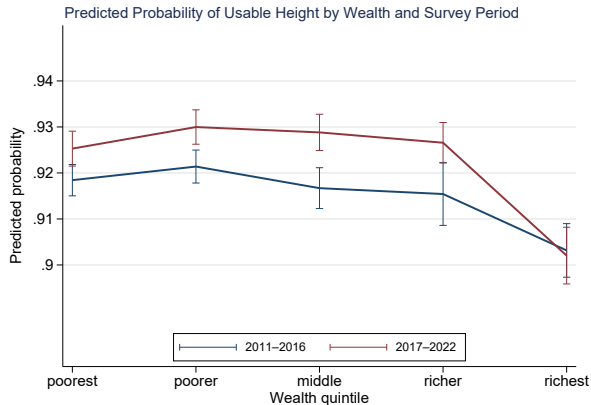


Result 4: Wealth pattern persisted over time

- Predicted usability was slightly higher in 2017–2022 than in 2011–2016
- The wealth gradient remained non-monotonic in both periods
- The richest quintile consistently had the lowest predicted usability

Why this matters

The result adds depth to the main model by showing that the unusual wealth pattern is persistent rather than temporary.



Result 5: Complete-case and MI HAZ results were broadly similar

- HAZ coefficients were highly similar under complete-case and multiple imputation models.
- Stability held across child, maternal, and household predictors.
- Substantive interpretation did not materially change after imputation.

Selected coefficient comparison

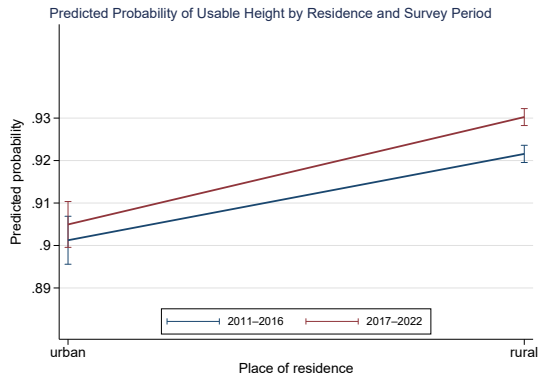
Predictor	CC	MI	Conclusion
Child age	-0.016	-0.016	unchanged
Female child	0.080	0.080	unchanged
Child illness	-0.079	-0.075	very small change
Children under five in HH	-0.086	-0.085	unchanged
Secondary education	0.303	0.303	unchanged
Higher education	0.533	0.532	unchanged
Currently pregnant	-0.235	-0.217	small change
Richest wealth	0.682	0.675	very small change

Interpretation

Systematic missingness does not automatically imply large inferential distortion in adjusted HAZ models.

Supplementary Analyses

Temporal variation from period-stratified adjusted models



- Child age remained positively associated with usable height in both periods.
- Illness and rural residence were also consistently associated with higher usability.
- Most coefficients were broadly stable across periods.
- Improvement in usability was modest rather than structural.

Takeaway

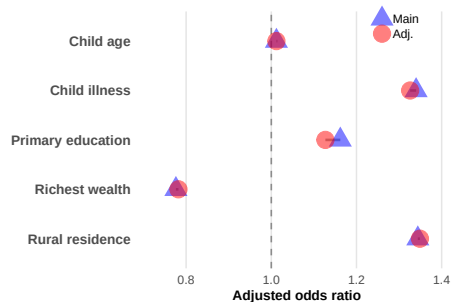
Temporal variation did not overturn the main conclusion that height usability is systematically patterned.

Robustness across alternative specifications

Predictor	Main	Excl. Age	Conclusion
Primary edu.	1.162	1.171	modest change
Secondary edu.	1.164	1.677	clear increase
Pregnant	1.169	0.806	direction reversed
Richest wealth	0.776	0.833	pattern retained
Rural	1.344	1.215	attenuated
Children under-5	0.966	1.175	direction reversed

Sensitivity analysis

Excluding child age shifted some coefficients, but the overall pattern remained.



Adjusted pooled model

Country and period adjustment did not materially change the main pattern

Overall takeaway

The main interpretation remained stable across alternative specifications.

Conclusion

- Anthropometric usability in pooled South Asian DHS data is structured rather than random.
- Younger children were most likely to have unusable height measurements, with additional patterning by education, wealth, and residence.
- Usability improved slightly over time, but the unusual wealth pattern persisted across survey periods.
- Complete-case and multiple imputation HAZ models produced broadly similar substantive conclusions.
- Missing-data diagnostics should therefore be treated as a standard part of secondary analysis using DHS anthropometric data.

Limitations and Future Directions

- Repeated cross-sectional DHS data allow identification of systematic associations, but not the causal mechanisms behind unusable anthropometric measurements.
- Pooling eleven survey rounds across four countries may leave residual heterogeneity despite harmonization and survey-design adjustment.
- Missing child age required sample restrictions, so analytic sample sizes differ across descriptive summaries, usability models, and HAZ regressions.
- Multiple imputation was implemented under a MAR framework; future work could extend country-specific diagnostics and formal sensitivity analysis under alternative missing-data mechanisms.

Key References

- DHS Program. Demographic and Health Surveys (2011–2022), Bangladesh, India, Pakistan, and Nepal.
- WHO Multicentre Growth Reference Study Group. WHO Child Growth Standards.
- Finaret, A. B., & Hutchinson, M. S. (2018). The missingness of DHS height data in Sub-Saharan Africa was non-random but unlikely to bias stunting estimates substantially. *The Journal of Nutrition*, 148(5), 781–789.
- Pullum, T. W. (2013). Anthropometric data in DHS surveys: issues and guidance.
- Rubin, D. B. (1987). *Multiple Imputation for Nonresponse in Surveys*.
- Ulijaszek, S. J., & Kerr, D. A. (1999). Anthropometric measurement error and the assessment of nutritional status.

Additional Interactive Dashboard

Dashboard link / QR placeholder

<https://tahsina-akter28.shinyapps.io/dashboard/>



Thank You!